SPARKY'S SCIENCE FACTS

How Do Plants Make Their Food?

Biology · Ages 7+

> **Short answer:** Plants make their own food in their leaves, using sunlight, water and air. This is called photosynthesis.

How it works

- Roots take in water from the soil. Tiny holes under the leaves take in a gas from the air called carbon dioxide.
- The green colour in leaves, chlorophyll, catches sunlight.
- The leaf uses the sunlight's energy to turn water and carbon dioxide into sugar. The sugar is the plant's food, for growing.
- The plant gives out oxygen, the gas we need to breathe.

Key words: Photosynthesis: making food from light; Chlorophyll: the green stuff in leaves

Common mix-ups

- Plants do not eat soil. Soil gives them water and minerals, but they make their food themselves.
- Plants need light to make food: a plant kept in a dark cupboard turns pale and weak.

Did you know? The Cape Floral Kingdom in the Western Cape has about 9,000 kinds of plants, and most of them grow nowhere else on Earth.

Try it: Cover one leaf of a pot plant with foil for a week (with a grown-up's OK). It turns pale without light.

Ask Sparky: Why are most leaves green? Do plants breathe?